Quantum Theory Needs Its Complex Numbers: Gravity and the Proposed Hidden Sector

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Abstract

Quantum theory needs its complex numbers under its standard rule for combining systems: real-number versions then predict different correlations in network experiments, and the experiments sided with the complex theory. On that rule the imaginary unit describes the world and is no bookkeeping shortcut; a 2026 preprint argues that a different reading of independence lets real versions match. That makes it natural to ask whether the imaginary part carries a hidden sector, mass that gravitates without shining. As a reading of the notation it cannot, because rotating the phase mixes the real and imaginary parts and changes no observation, so neither part alone can be the dark one. A complex field that is neutral under electromagnetism can carry one, though: such a field gravitates while staying dark, and in the right mass range it behaves like cold dark matter on cosmological scales, with the darkness belonging to the whole neutral field. The paper works out the minimal model, what would make one component physically preferred, which astronomical observations would decide, and how complex variables enter gravity and singularities.

1. What the imaginary unit establishes

Complex numbers in quantum theory do not by themselves identify a hidden sector, unseen matter that still exerts gravity. They have real and imaginary components (the imaginary one multiplies the square root of minus one), and

quantum theory uses them to describe how contributions combine and interfere. Calling a component imaginary does not mean it is an unseen material. The question is whether a physical model can turn that mathematical distinction into an explanation of gravity or dark matter.

Can a complex quantity be written as a pair of real quantities? Algebraically, yes. Does a particular theory on real Hilbert spaces (quantum state spaces using only real numbers) reproduce the same composition (how systems combine), preparation and measurement structure as standard complex quantum theory? That requires analysis of the complete theory.

Calling complex numbers mere notation can miss a physically consequential structure. Calling their use proof of a hidden imaginary mass sector goes beyond the same evidence. The proposal must identify which transformations and observables (measurable quantities) have physical significance before assigning separate ontological roles (different statuses in what really exists) to the two coordinate components.

2. Complex variables in gravitational theory

Complex variables do real work in gravitational theory without supplying a map between gravity and quantum theory. General relativity uses a real Lorentzian metric, a spacetime distance rule treating time differently from space. Quantum states are represented in complex Hilbert spaces. This contrast can motivate a search for common structures, but it is not a proof that their incompatibilities reduce to choosing different number systems.

Roger Penrose's twistor theory restates aspects of spacetime and field theory in the language of complex geometry, trading spacetime points for the points of a complex space of its own (Penrose, 1967). Abhay Ashtekar rewrote gravity using a complex connection, a variable saying how to compare directions at neighboring

points, and the gravitational constraints became simpler; extra requirements, called reality conditions, then pick out the real Lorentzian physics the theory is meant to describe (Ashtekar, 1986). Those conditions are part of specifying the physical theory. Edward Witten's discussion of allowable complex spacetime metrics concerns which complex configurations can be admitted under specified conditions (Witten, 2021). It does not establish that every complex metric is a physical spacetime.

These approaches make complex structure worth taking seriously. They do not yet provide a common map between their variables, state spaces, observables and dynamics. Similar-looking complex notation does not ensure that two theories assign the same physical meaning to it.

2.1 What a common complex geometry would have to unify

A successful unification should specify three maps: recover an observable Lorentzian metric with the intended signature (one time and three space directions); identify the quantum state space, composition rules and observables; and recover the relevant dynamics including the interaction between matter and geometry. A formal map at only one level is not a solution to the other two. Reality conditions are part of the first map, not evidence that the physical theory has artificially discarded a dark half of itself.

Complex structures keep proving useful, which may indicate that the right description at bottom makes phase (the angle of a complex number drawn as an arrow) and geometry aspects of one construction. The supporting evidence consists of the relations each program has actually demonstrated, not independent confirmation of an unspecified real-visible/imaginary-dark projection.

3. The phase test for a hidden imaginary sector

Rotating the phase mixes the real and imaginary components, so any claim that one of them is the dark half has to survive that rotation. In symbols, let a complex field be Phi = a + ib. A phase rotation through theta gives

$$a' = a\cos\theta - b\sin\theta, \qquad b' = a\sin\theta + b\cos\theta.$$

The modulus, the size of the complex value, is invariant; the separate components are not. A field written as purely imaginary in one phase convention can be purely real in another.

If such a phase change is a gauge redundancy, a relabeling with no physical content, observable darkness cannot depend on which description is selected. If it is a physical global symmetry (the same transformation applied at every point at once), a proposal must state what interaction or symmetry-breaking structure selects one component for a different coupling, a distinct way of interacting with other fields. A preferred real-axis detector is additional dynamics, not a consequence of complex notation.

Electromagnetic coupling reaches both components at once. Written out, for a charged complex scalar, a field with one complex value at each point, with D_mu = partial_mu + iq A_mu,

$$|D_\mu \Phi|^2 = (\partial_\mu a - qA_\mu b)^2 + (\partial_\mu b + qA_\mu a)^2.$$

Both components enter the charged-field interaction. If the field carries no electromagnetic charge, both can be electromagnetically dark. Darkness is determined by the couplings and representation (how the field transforms under symmetries), not by the word imaginary.

This is the central obstacle to the imaginary-sector interpretation. Transform the fields according to the proposed theory and check whether the visible-dark distinction changes under a transformation that should leave observations unchanged.

4. Scalar-field dark matter and what it supports

Scalar dark matter models exist without a literal imaginary component. Ultralight scalar-field dark matter is an established model class, with wave effects on astrophysical scales for appropriate masses (Hui et al., 2017). Complex models carry a conserved charge under global U(1) symmetry, invariance under rotating the field's phase. Real scalar models also exist; wave behavior is not exclusive to a literal imaginary component.

A model requires a field action (the quantity from which the field's equations follow), mass, interactions, cosmological preparation and a calculation of observables. Solitonic-core behavior, a stable dense clump of the field at a galaxy's center, in simulations (Schive, Chiueh and Broadhurst, 2014) concerns specified models. It does not independently confirm that reality contains a gravitationally visible imaginary axis.

4.1 A minimal invariant model

A complete action shows darkness belonging to the whole neutral field. Take a complex scalar Phi neutral under electromagnetism. Written out, in units with hbar = c = 1 and signature (-,+,+,+),

$$S_\Phi = \int d^4x \sqrt{-g} \left[-g^{\mu\nu} \partial_\mu \Phi^* \partial_\nu \Phi - m^2 |\Phi|^2 - \frac{\lambda}{2} |\Phi|^4 \right].$$

Here m^2 > 0 and lambda >= 0. The action has a global U(1) phase symmetry. Varying the metric gives a stress-energy (the field's energy, momentum and pressure) that is phase invariant and can source gravity. Stated precisely,

$$T_{\mu\nu} = \partial_\mu \Phi^* \partial_\nu \Phi + \partial_\nu \Phi^* \partial_\mu \Phi - g_{\mu\nu} \left(g^{\alpha\beta} \partial_\alpha \Phi^* \partial_\beta \Phi + V \right).$$

In the quadratic, rapidly oscillating regime with m much larger than the expansion rate, the averaged kinetic and potential contributions balance and the mean pressure approaches zero. A scalar can behave like cold matter at the background level, the smoothed-out average of the expanding universe.

The example establishes: a complex field can gravitate while being electromagnetically dark. It does not establish that its imaginary part alone is dark. The invariant action makes the opposite point: the darkness belongs to the whole neutral field.

4.2 How a physically preferred component could arise

A mass splitting earns the distinction that notation alone cannot. In symbols, write Phi = (a + ib)/sqrt(2) and add Delta V = epsilon(Phi^2 + Phi*^2)/2. The quadratic potential becomes

$$V_2 = \frac{1}{2}(m^2 + \varepsilon)a^2 + \frac{1}{2}(m^2 - \varepsilon)b^2.$$

For |epsilon| < m^2 both squared masses are positive. The U(1) symmetry is broken to a discrete symmetry (one with separate steps, such as a sign flip) and the two mass eigenmodes (field combinations with definite masses) have different dynamical roles. Renaming coordinates cannot erase the physical mass splitting. This shows what it means to earn a component distinction.

It still does not assign electromagnetic visibility to one mode. That requires specified gauge representations, mixing or further interactions. The same construction cannot simply be added to an electromagnetically charged scalar: Phi^2 would generally violate its gauge symmetry without additional charged structure.

4.3 Which astronomical claims follow

Solitonic cores, wave interference and small-scale suppression arise for particular field masses and dynamics, not for every complex field. The observational program must derive a background history, perturbation transfer function (how early density ripples grow or fade at each scale) and halo behavior (how the dark matter clouds around galaxies behave), then confront the relevant data. Bounds from the Lyman-alpha forest, hydrogen absorption lines in the light of distant quasars, constrain modeled suppression; they do not measure an imaginary axis. The program gains credibility by keeping the theoretical prediction, the observation and the inference between them separate.

5. Complexification and singularities

A complex extension does not generally remove a singularity, a point where quantities become infinite or undefined. The function 1/z still has a pole, a point where its value blows up, at z = 0. A contour, a path through the plane of complex numbers, can avoid the pole without making the function regular at the excluded point.

General-relativistic singularity theorems concern geodesic incompleteness under specified assumptions: a freely falling path that simply stops after finite proper time, the time a clock on that path would record. A proposed resolution must identify the relevant spacetime, geodesics and physical continuation rather than demonstrate only that a complex contour can be drawn. The Hartle-Hawking no-boundary proposal, the idea that the universe has no boundary at its beginning, illustrates a substantive use of complex methods in quantum cosmology (Hartle and Hawking, 1983). It is not a theorem that every singular Lorentzian solution becomes regular through analytic continuation, extending a function uniquely beyond the region where it was first defined.

A candidate resolution should establish what replaces the classical description, which observables remain finite and whether evolution can be continued. Complexification is a tool with potentially important consequences, not a universal removal operation.

6. Real and complex quantum theories

Whether real numbers suffice depends on the composition and independence rules assumed. Renou and colleagues studied network scenarios in which real and complex quantum models give different correlations under specified source-independence and composition assumptions (Renou et al., 2021). Chen and colleagues performed a related experimental test (Chen et al., 2022). Hoffreumon and Woods (2026) argue that once independence is formulated operationally, real models reproduce complex-theory correlations more broadly.

The composition and independence rules matter. An experiment can distinguish theories sharing some algebraic ingredients while differing in those rules. These findings support careful attention to complex structure without establishing the imaginary-sector dark-matter proposal, which requires its own coupling mechanism.

7. Phase, amplitude and measurement

In the exponential, one part sets a phase and the other sets a magnitude. Written out, for a complex action S = A + iB, exp(iS/hbar) factors into exp(iA/hbar) times exp(-B/hbar). The real and imaginary parts have distinct roles: phase and magnitude. That identity can guide a model, but it does not show that the classical action is in every case the real projection of a physical complex action.

A measurement that returns a real number does not thereby measure only the real part of a wavefunction, the mathematical description of a quantum state. Quantum expectation values (average results over repeated measurements) of Hermitian observables (those whose measured values are real numbers) are real even though they depend on the complex state and its phase relations.

8. A program with discriminating tests

The four questions, gravitational unification, a dark sector, singularities and the quantum formalism, have separate completion conditions. A gravitational unification must define a common structure and recover the relevant dynamics. A dark-sector model must specify its invariant visible-dark distinction and predict observables. A singularity proposal must state what is continued and under which criteria. A quantum-formalism proposal must identify its composition assumptions and their observational implications.

The negative checks are equally useful. A gauge-dependent darkness criterion fails. A complex pole remains a pole. A real-coordinate rewriting that preserves all operational predictions is not experimentally rejected by its notation. These checks locate where interpretation must become physical construction.

The neutral-scalar example answers the first constructive question: a consistent action can exhibit electromagnetic darkness and gravitational sourcing, but it does not implement the literal imaginary-sector thesis. The split-mass example answers a second: a component distinction becomes physical when an interaction defines it. If the same parameters control more than one successful prediction, the unification acquires explanatory force. If each question requires unrelated new assumptions, complex notation alone has not supplied the promised economy.

9. Conclusion

Complex numbers earn their place in quantum theory by describing how contributions combine and interfere. They do not thereby divide nature into visible matter and a hidden gravitational sector. The calculations here show what such a division would require: a physical reason for telling the components apart, equations governing each, and an observation that comes out differently as a result. Mathematical monism, the view that reality is mathematical structure, is a reason to look for one common account. It is not a reason to accept any particular construction before that construction shows what it explains.

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